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Rules versus Enforcement: Stablecoin Peg Stability Around MiCA and Binance's Delisting Announcement.

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Abstract

Stablecoins have become an important part of digital-asset markets, with their market capitalization increasing sharply since 2020 from USD 5 billion to USD 323 billion as of May 2026. Their usefulness depends on their ability to remain close to their intended peg which is usually 1 USD. Deviations from the one-dollar peg can indicate pressure on market confidence and liquidity and the ability of stablecoins to function as trading pairs, settlement instruments and liquidity bridges. This makes the regulation of stablecoins important not only as a legal issue but also a potential source of market instability. The European Markets in Crypto-Assets Regulation introduced a new regulatory distinction between compliant and non-compliant stablecoins, and major exchanges were required to adjust the availability of affected assets for European Economic Area users. This thesis examines whether MiCA-related regulatory developments affected the short-run peg stability of USD-pegged stablecoins on Binance. The analysis focuses on Binance hourly price data and compares stablecoins affected by MiCA-related restrictions with USDC, which serves as the MiCA-compliant control stablecoin. Treated stablecoins include FDUSD, TUSD, and USDP. Absolute peg deviation is calculated as the absolute difference between each stablecoin's hourly price and its intended one-dollar peg. I estimate difference-in-differences models around two events: the broad MiCA stablecoin-rule application date on 30 June 2024 and Binance's exchange-specific restriction announcement on 3 March 2025. The results show that the broad MiCA application date does not produce a statistically significant increase in peg deviations. In contrast, Binance's 3 March 2025 announcement is associated with a positive and statistically significant increase in peg deviations for treated stablecoins relative to USDC. The estimated effect is approximately 4.34 basis points. Robustness checks show that the effect is similar for both high-volume and low-volume treated stablecoins. Overall, the findings suggest that exchange-specific MiCA-related restrictions had a stronger short-run effect on stablecoin peg stability than the broad regulatory application date itself.

1. Introduction

This thesis examines whether MiCA-related restrictions on non-compliant stablecoins increased short-run peg deviations on Binance. The central research question is: Do Binance's MiCA-related stablecoin restrictions increase short-run peg deviations for affected USD-pegged stablecoins relative to the MiCA-compliant stablecoin USDC? This question is important because MiCA changed the regulatory conditions under which stablecoins can be offered to EEA users. If exchange-level restrictions reduce the liquidity or usability of affected stablecoins, these restrictions may create temporary pressure on their ability to trade close to one dollar which can lead to loss of confidence among the users who often use stablecoins as a collateral in crypto markets and cause liquidations. Market makers may reduce trading activity or widen spreads if they are unsure whether the stablecoin will return to one dollar.

Stablecoins have expanded rapidly in recent years. Ahmed, Aldasoro and Duley (2025) report that total stablecoin market capitalization increased from around USD 5 billion in early 2020 to USD 190 billion in May 2022 and USD 203 billion by December 2024. More recent estimates place the market above USD 300 billion in 2026, it is already comparable in scale to large individual institutional investors. For example, ABP, the largest Dutch pension fund, reported EUR relevant for digital533 billion in available assets at year-end 2025. This scale makes stablecoin peg stability -asset markets, where stablecoins are widely used as trading pairs, settlement instruments, collateral, and liquidity bridges.

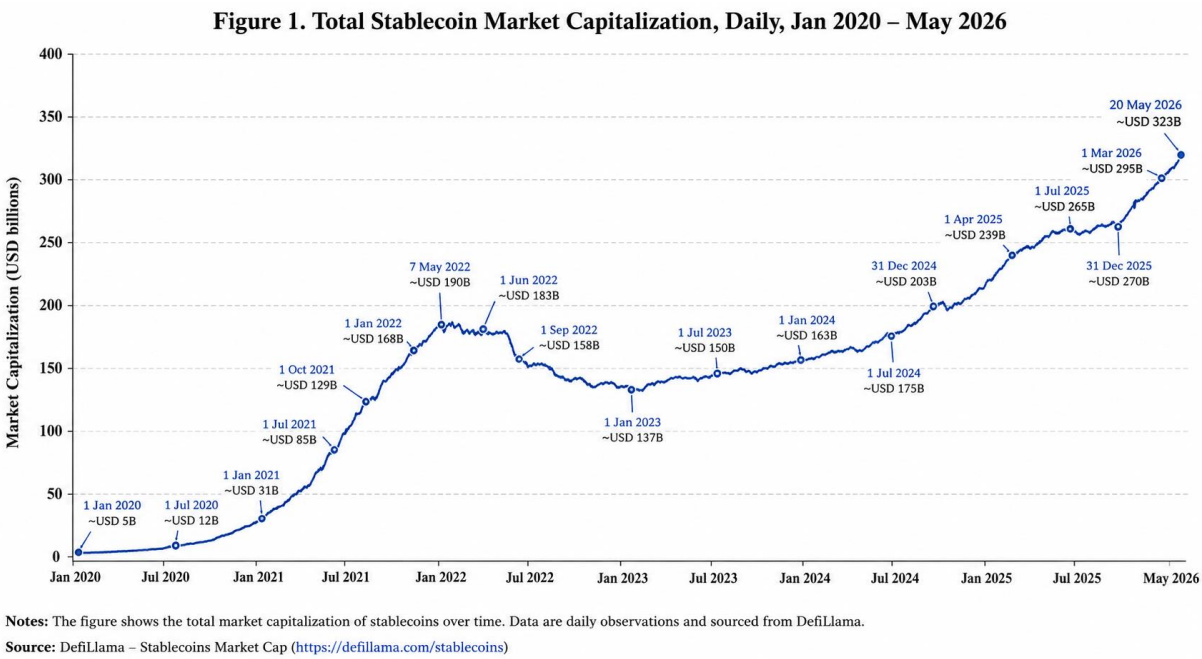


Figure 1. Growth of Global Stablecoin Market Capitalization

Notes: *This figure shows selected benchmark points in the growth of global stablecoin market capitalization. Stablecoin market capitalization increased from approximately USD 5 billion in early January 2020 to USD 190 billion in early May 2022, before reaching USD 203 billion in December 2024 and more than USD 270 billion by December 2025. The latest DefiLlama figure reports total stablecoin market capitalization of approximately USD 323 billion.*

Recent studies show that stablecoin pegs are not automatically guaranteed to be stable. They are designed to maintain a stable value relative to a reference asset, most commonly the U.S. dollar. However, this stability depends on market confidence, liquidity, reserve quality, and the ability of traders and arbitragers to correct price deviations. Ahmed, Aldasoro and Duley (2025) show that public information about reserve quality, transparency, and reserve-asset volatility can affect stablecoin peg stability. In other words, if market participants receive negative information about whether a stablecoin is safely backed or easily redeemable, the stablecoin may trade below its intended peg.

A clear example is the Silicon Valley Bank crisis in March 2023. Silicon Valley Bank was a U.S. bank that failed after concerns about its financial position led to a rapid withdrawal of deposits. Circle, the issuer of the USD Coin stablecoin (USDC), disclosed that USD 3.3 billion of its cash reserves were held

at Silicon Valley Bank. Because USDC is designed to maintain a one-dollar value, this information created concern that part of the reserves backing USDC might not be immediately available. As a result, some holders sold USDC, and the price temporarily fell below one dollar. This episode illustrates how public information can affect confidence and create de-pegging pressure even when a stablecoin is designed to remain stable (Ahmed, Aldasoro and Duley, 2025).

Stablecoin stability also depends on arbitrage and market structure. Stablecoin arbitrage refers to the process through which traders help restore the peg when a stablecoin trades away from one dollar. For example, if a stablecoin trades below one dollar, arbitragers may buy it at a discount and either redeem it or sell it later closer to par. This buying pressure should push the market price back toward one dollar. However, arbitrage does not work equally well for all stablecoins. Ma, Zeng and Zhang (2025) show that redemption-based arbitrage is highly concentrated and differs across issuers. This suggests that peg stability depends not only on reserves, but also on whether arbitragers and market makers can absorb buying and selling pressure quickly.

The empirical setting is Binance, because it is the largest crypto exchange in the world by trading volume and hence a dominant venue where stablecoins are traded. USDC is the control coin, which is the second largest stablecoin with a market share of 25% behind tether. In this thesis, exchange restrictions refer to measures taken by crypto-asset trading platforms to limit or remove access to stablecoins that are not compliant with MiCA requirements for EEA users. Under MiCA, issuers of asset-referenced tokens and e-money tokens must obtain the relevant authorization to offer these tokens in the EU. Crypto-asset service providers, including exchanges, also operate under MiCA and are expected to stop offering or supporting stablecoins that do not meet MiCA requirements. In practice, this means that authorised exchanges may restrict, convert, or delist non-MiCA-compliant stablecoin trading pairs for EEA clients. Stablecoins can be grouped by mechanism of how they are backed namely backed by collateral reserves, by cryptocurrency and some rely on algorithmic supply adjustments. The treated and control coins in this thesis are all backed by collateral or fiat reserves.

This mechanism is relevant for peg stability because exchange restrictions can reduce a stablecoin's accessibility and trading liquidity. If users expect that a stablecoin will become harder to trade or less useful on major exchanges, they may shift toward compliant alternatives such as USDC. This reallocation can create short-run buying or selling pressure and may cause affected stablecoins to move further away from their intended one-dollar peg. Evidence from the BUSD delisting shows that regulatory actions can reshape stablecoin market dynamics. Diop and Chevallier (2026) find that liquidity displaced from BUSD was reallocated mainly toward USDT and USDC, increasing stablecoin market concentration and highlighting a trade-off between regulatory enforcement and market structure. Similarly, MiCA-related restrictions may shift liquidity away from non-compliant stablecoins and toward compliant alternatives, potentially affecting short-run peg deviations.

To answer the research question, I use hourly Binance price data for USD-pegged stablecoins. On Binance these coins are quoted against USDT rather than fiat US dollars, so USDT serves as the dollar proxy, and peg deviation is measured relative to a one-to-one rate with USDT. The treated group consists of the stablecoins affected by Binance's MiCA-related restrictions — FDUSD, TUSD, and USDP. USDC is used as the control stablecoin because it remained available as a MiCA-compliant alternative. The main dependent variable is absolute peg deviation, defined as the absolute difference between a stablecoin's hourly price and its intended one-dollar peg. The empirical analysis uses a difference-in-differences design to compare changes in peg deviations for affected stablecoins relative to USDC around MiCA-related events. I first examine the broad MiCA stablecoin-rule application date of 30 June 2024 and then focus on Binance's exchange-specific announcement on 3 March 2025.

This thesis contributes to the literature by linking stablecoin peg stability to regulatory exchange restrictions. Existing research has examined public information shocks, reserve disclosures, arbitrage concentration, and earlier delisting events such as BUSD. However, less is known about whether MiCA-related restrictions created measurable short-run effects on the peg stability of affected

stablecoins in the EEA. By using hourly exchange-level data from Binance, this thesis provides evidence on whether regulatory restrictions affected stablecoin peg deviations at the point where market access changed.

The rest of this thesis is structured as follows. Section 2 reviews the literature on stablecoin peg stability, public information shocks, arbitrage, and MiCA-related regulation. Section 3 describes the data and variable construction. Section 4 presents the empirical methodology. Section 5 reports the main results and robustness checks. Section 6 discusses the limitations and future research. Section 7 concludes.

2. Literature Review

This thesis adds to a growing literature on stablecoins and determinants of their peg stability. The stability of stablecoin to trade near one US dollar is not automatic, several research papers show that it is a combination of sufficient liquidity, confidence in the issuer, quality and accessibility of reserves, capacity of arbitragers to correct deviations from par. The literature reviewed below can be read as a single account of how stablecoins are held in place, why they nonetheless break, and how regulatory interventions affect the market conditions in which these stablecoins operate.

The first and most fundamental mechanism through which a stablecoin maintains its peg is arbitrage. In principle, when a fully-backed coin trades below one dollar, participants able to redeem it for a dollar of reserves, or to buy it cheaply and sell it elsewhere closer to par, profit by doing so, and the resulting buying pressure pushes the price back toward parity; symmetric pressure operates when the coin trades at a premium. The implication is that deviations should be small and short-lived

wherever arbitrage is efficient and liquidity is deep. Lyons and Viswanath-Natraj (2023) formalise and test this view, showing that the strength and accessibility of the arbitrage channel is a primary determinant of how tightly a stablecoin tracks its peg. Using a difference-in-differences design that compares Tether to a control group of other stablecoins which did not undergo the migration, they find that the improvement in arbitrage access reduced the absolute size of Tether's peg deviations by roughly a third relative to the control group. They further show that this stabilising effect operates through arbitrage flows between investors and the issuer. Empirical work on Tether's price dynamics complements this by linking the stabilising force directly to market conditions, finding that mean reversion to the peg is stronger when the stablecoin market is more liquid, and that Tether's role as a safe haven during periods of elevated Bitcoin volatility further supports its price (Hui, Wong and Lo) (2025). Together these studies establish that a peg is not held by design alone but by the active participation of arbitragers operating in liquid markets.

A crucial qualification to this mechanism is that arbitrage does not operate equally across issuers. Ma, Zeng and Zhang (2025) examine the structure of redemption-based arbitrage and find it to be strikingly concentrated: in an average month, USDT was redeemed by only six arbitragers, compared with 521 for USDC. They further show that coins served by fewer arbitragers exhibit larger secondary-market deviations from par. This finding refines the simple arbitrage view in an important way: peg stability depends not merely on whether a redemption mechanism exists, but on whether enough participants can actually access it to absorb short-run buying or selling pressure. The practical relevance for the present study is direct. An exchange delisting is precisely an event that reduces the set of venues and participants through which a coin can be traded and arbitraged, so the framework of Ma, Zeng and Zhang (2025) predicts that such a restriction should weaken the stabilising mechanism, and that the effect may be larger for coins whose arbitrage and liquidity are

more concentrated or more exchange-dependent. This motivates the present study's examination of whether the effect of the Binance restriction differs across treated coins of differing trading volume.

If the first group of studies explains what holds pegs in place, a second group documents that pegs nonetheless deviate, and that these deviations are driven by information and market conditions and are transmitted across coins rather than occurring in isolation. Ahmed, Aldasoro and Duley (2025) show that public information about reserve quality, transparency, and reserve-asset volatility can shift whether users continue to trust a peg. Their central insight is that instability can arise well before a stablecoin is fundamentally insolvent: if holders come to believe that others may redeem or sell, it becomes individually rational to sell first, producing a coordination problem in which negative information alone generates short-run selling pressure and a discount to par. The silicon-valley episode of March 2023 demonstrates that even a large, transparent, fully-collateralised stablecoin can lose its peg in the short run when confidence in the accessibility of its backing is disturbed. The present study builds on this insight by treating Binance's MiCA-related restriction as a different species of information shock: it did not alter the reserves backing the affected coins, but it changed expectations about whether those coins would remain tradable and useful for EEA users, and may therefore have influenced confidence and short-run peg stability.

Beyond issuer-specific information, the broader market environment also drives depegging, and the relevant deviations are interconnected across coins. The interdependence of stablecoins is made central in work on co-instability, which shows that the leading coins do not move independently but exhibit strong co-movement and correlated peg dynamics, with spillovers running mainly from collateralised toward algorithmic coins (Gadzinski et al., 2024). De Blasis et al. (2023) provide direct evidence of this transmission around the May 2022 collapse of TerraUSD. Using intraday data in an event-study framework, they document contagion across stablecoins and crypto-assets and show that the magnitude, direction, and duration of each coin's response depended on its underlying design i.e. redemption mechanism, fiat/cryptocurrency/algorithmic, dependence on another token.

During the same episode, some coins traded at a discount while others rose to a premium. This body of work has two implications for the present study's design. First, because stablecoins respond jointly to common shocks, a difference-in-differences specification in which a compliant control coin absorbs market-wide movements is the appropriate way to isolate the effect of a coin-specific event. Second, because instability is transmitted across coins, it cannot be assumed that treated and control coins were on parallel paths before the event, which makes an explicit pre-trend test essential rather than optional.

The third and most closely related body of work concerns how regulation, and the delistings it induces, reshape stablecoin markets. The EU's Markets in Crypto-Assets regulation classifies stablecoins as electronic-money or asset-referenced tokens and subjects their issuers to authorisation, reserve, and governance requirements; critically, it places the compliance burden on service providers as well, so that an authorised exchange risks its own standing if it continues to offer non-compliant tokens to EEA users (Vidal-Tomás, 2025). Santner and Taudes (2025) draw out the market consequence of this design, arguing that MiCA turns regulatory compliance into a determinant of market access: instability can arise not only from reserve concerns or market panic, but from regulatory restrictions that reduce a coin's usability and liquidity. In their framework, if users anticipate that a non-compliant coin will become harder to trade, they rationally shift toward compliant alternatives such as USDC, and this migration can itself generate short-run pressure that widens the affected coins' deviations. Diop and Chevallier (2026) provide direct empirical evidence of such reallocation in a closely analogous setting, studying the delisting of Binance USD and finds that the liquidity displaced from the delisted coin flowed mainly toward USDT and USDC, raising concentration in the stablecoin market. Their result establishes that an exchange-level delisting affects not only the targeted coin but the broader allocation of activity across the stablecoin market.

Taken together, these three bodies of work leave a specific gap that this thesis addresses. The first establishes that peg stability depends on arbitrage access and liquidity; the second shows that pegs

break under information and market shocks and that instability is transmitted across coins; the third shows that regulatory delistings reduce market access and reallocate liquidity toward compliant coins. Yet no study causally identifies the effect of a regulatory-driven delisting on peg stability itself. This thesis fills that gap by using hourly, exchange-level Binance data and a difference-in-differences event study to isolate the short-run effect of a MiCA-related delisting on absolute peg deviation. The institutional setting and the construction of this design are set out in the following sections.

3. Data

3.1 Empirical Setting

Under MiCA's Title V part of Regulation (EU) 2023/1114, applicable from 30 December 2024, crypto-asset service providers operating in the EEA are prohibited from offering stablecoins whose issuers have not obtained the required authorisation. Exchanges serving EEA users therefore faced a binary choice: remove non-compliant stablecoins or jeopardise their own ability to operate legally in the region. In practice, this meant that authorised or prospective service providers had to restrict, convert, or delist non-MiCA-compliant stablecoin pairs for EEA clients.

This thesis uses Binance as its empirical setting. Binance is the largest cryptocurrency exchange in the world by trading volume, and therefore the dominant venue on which the affected stablecoins are priced and traded, which makes it the most economically significant setting in which to observe a MiCA-driven delisting. Binance's own position under MiCA is central to the design: during the sample period it was operating under MiCA's transitional arrangements while pursuing full authorisation, which it obtained in 2025, and it acted to align with the regulation by restricting and delisting non-compliant stablecoin pairs for EEA users. This restriction was announced on 3 March 2025 and took effect on 31 March 2025. The control coin, USDC, remained available throughout precisely because its issuer had obtained MiCA authorisation. Beyond its economic significance, Binance also offers a

clean empirical setting: it provides a single, clearly identifiable announcement date and hourly exchange-level price data for both the affected stablecoins and the compliant control, which offers sharper identification than a market-wide analysis that would combine multiple exchanges with differing announcement dates, affected coins, and implementation timelines, and would measure prices away from the venue where the restriction occurred. Stablecoins can be grouped by the mechanism through which they maintain their peg. Fiat-collateralised stablecoins are backed by reserves of cash and cash-equivalent assets, such as bank deposits and short-term government securities, held by a centralised issuer and intended to support redemption at close to one dollar; their stability depends on the quality, liquidity, and transparency of those reserves. Crypto-collateralised stablecoins, such as DAI, are instead backed by other cryptocurrencies held as over-collateralised reserves, with the excess collateral absorbing the volatility of the backing assets. Algorithmic stablecoins attempt to hold the peg without full collateral, adjusting supply through incentives and a linked token; this design is the most fragile, as illustrated by the May 2022 collapse of TerraUSD, in which the mechanism failed to defend the peg and the coin lost its value entirely. The stablecoins examined in this thesis FDUSD, TUSD, and USDP among the treated coins, and USDC as the control are all fiat-collateralised US-dollar stablecoins. Because they share the same backing mechanism, differences in their behaviour around the events studied are not attributable to fundamentally different stabilisation designs.

The literature suggests that such restrictions may affect peg stability through three related channels. First, the restriction can act as a public information shock, changing users' expectations about whether a stablecoin will remain accessible and usable in the EEA. Second, it can reduce liquidity by limiting the venues and trading pairs through which affected coins can be bought and sold. Third, it may encourage users to shift from non-compliant coins toward compliant alternatives such as USDC; if this reallocation creates short-run pressure that arbitrageurs and market makers do not fully absorb, the affected coins may move further from their one-dollar peg.

The empirical analysis examines two MiCA-related events. The first is the broad MiCA stablecoin-rule application date of 30 June 2024, which marks the general regulatory milestone at which the rules for stablecoin issuers became applicable. Because this date was known well in advance and did not correspond to any specific Binance trading restriction, the expected short-run market reaction is limited, and it therefore serves as a comparison event. The second is Binance's announcement of 3 March 2025, which informed EEA users of restrictions on non-compliant stablecoins; being directly tied to exchange-level market access, it is expected to generate a clearer short-run response in peg deviations. The contrast is informative: if the broad regulatory date produces no significant effect while the exchange-specific announcement does, this would suggest that short-run peg pressure arises when regulation becomes operational through concrete exchange-level restrictions rather than at the broad regulatory milestone. These considerations motivate the central hypothesis, which is stated formally alongside the empirical specification in the methodology section.

Finally, this thesis examines whether the effect differs by trading volume. The liquidity literature suggests that lower-liquidity stablecoins may be more sensitive to restrictions because their prices are harder to stabilize when trading pressure changes. To assess this mechanism, I conduct robustness checks that split treated stablecoins into high-volume and low-volume groups and estimate whether the treatment effect varies with pre-event trading volume. This allows the analysis to test whether the estimated effect is driven mainly by less liquid stablecoins or whether it is present across treated stablecoins more broadly.

3.2 Data sourcing and construction

The analysis uses hourly market data retrieved programmatically from the public Binance market-data interface (the Binance Spot REST API). For each stablecoin trading pair, the interface returns hourly candlestick (“Kline”) records, each comprising the open, high, low, and close prices and the traded volume within the hour. Data were collected for the period 1 June 2024 to 1 April 2025, a

span of 304 days, and stored as a single panel dataset. The central event under study is Binance's delisting announcement of 3 March 2025. Crucially, the affected coins continued to trade normally on Binance throughout the announcement window, so their prices over this period are formed on the same venue at which the restriction was announced. The subsequent effective delisting of 31 March, after which the coins ceased trading on Binance, cannot be analysed on the same basis, since post-delisting prices become stale; it is therefore treated as a separate, inconclusive event and discussed among the limitations. The availability of granular data was best available on Binance which made it obvious choice. The hourly frequency was chosen over the daily frequency used in much of the existing stablecoin literature for two reasons. First, it provides a more in-depth view required to define a tightly bounded event window and to trace the dynamic path of any effect within days of the event rather than only across them; at daily frequency a ±7-day window would contain only a handful of observations per coin, too few to estimate a short-run reaction. Second, hourly data makes it possible to absorb intraday structure directly: stablecoin liquidity follows time-of-day patterns that widen deviation at certain hours regardless of treatment, and hour-of-day fixed effects remove these patterns so that the effect is identified from within-hour comparisons. This is not feasible with daily data, which has no intraday variation to control for. Within each ±7-day estimation window the hourly panel provides 1,344 coin-hour observations across the four coins. Four US-dollar-pegged stablecoins are included. The treated group comprises the three coins that Binance delisted for European Economic Area users under MiCA and for which liquid Binance pairs existed throughout the sample: First Digital USD (FDUSD), TrueUSD (TUSD), Pax Dollar (USDP). The control is USD Coin (USDC), which remained supported on Binance for EEA users because its issuer obtained MiCA authorization.

The way these coins are quoted on Binance affects what the peg is measured against, so it must be made explicit. On Binance, FDUSD, TUSD, USDP and USDC are quoted against USDT and not against US dollars. Each coin’s price is therefore its value in USDT and its peg deviation is measured relative to a one-to-one rate with USDT. Thus, USDT serves as the dollar proxy throughout. USDT as the dollar

proxy is appropriate for three reasons. First, USDT is itself a fiat collateralised dollar stablecoin that trades close to one dollar under normal conditions, so the error introduced by treating it as a dollar reference is small. Second, on Binance USDT is the dominant unit of account as majority of pairs including those studied here are quoted against USDT rather than against US dollars. So it is the economically relevant benchmark against which these coins are actually priced on the venue. Third and most important point is the difference-in-differences design does not require USDT to be a perfect dollar as every coin is measured against the same USDT benchmark. Any movement in USDT will shift all of them together and is differenced out of the estimate. The treatment effect is identified from the relative movement between treated and control coins and not from their common exposure to USDT. The main caveat is that USDT can itself drift from one dollar in times of market stress, but because all coins are measured against the same USDT, any such movement affects them equally and does not distort the comparison between treated and control coins. Prior cryptocurrency market research supports this measurement choice (Wątorek et al., 2021).

The dependent variable is the absolute peg deviation, measuring how far its price sits from its one-dollar target irrespective of the direction. For coin *i* in hour *t* it is defined as the absolute difference between the coin's hourly closing price and one:

$$abs_peg_deviation = |close - 1|$$

In all regressions and tables this quantity is expressed in basis points, obtained by multiplying the price-unit deviation by 10,000, so that a price of 0.9990 or 1.0010 each corresponds to a deviation of ten basis points. The absolute value is used rather than the signed deviation because loss of peg credibility is associated with departures from par in either direction: a coin trading above its peg is off its peg just as one trading below it is. No closing prices are missing in the sample. In practice, since the coins are quoted against USDT on Binance, the one-dollar target is measured as a one-to-one rate with USDT, consistent with the use of USDT as the dollar proxy described above.

Hourly trading volume is retained for each coin-hour and is used in the robustness analysis to test whether the main effect varies with liquidity. Before estimation, the raw hourly prices were processed once into an analysis ready dataset. For each event, I added a set of variables used in the regressions: an event-time variable measuring how many hours each observation falls before or after the event, a corresponding day counter, and a post event indicator equal to one for hours after the event. A treatment indicator was set to one for the three delisted coins (FDUSD, TUSD, USDP) and zero for control (USDC), and the two were multiplied to form the treatment and post interaction that identifies the difference-in-differences effect. Calendar date and hour of the day were also recorded for use as fixed effects.

The market is highly concentrated: the two largest coins, USDT and USDC, together account for roughly two-thirds of total capitalisation. USDT, at about USD 142 billion (roughly 48% of the market), is the single largest stablecoin and is excluded from the analysis because it serves as the dollar benchmark against which the other coins are measured. The control coin, USDC, is the second-largest at about USD 56 billion (roughly 19% of the market). The treated coins are by contrast small: First Digital USD (FDUSD) at about USD 2.0 billion, TrueUSD (TUSD) at about USD 0.5 billion, and Pax Dollar (USDP) at about USD 0.08 billion, together representing under one percent of total stablecoin capitalisation. The sample is therefore not intended to be representative of the market by size; it is defined instead by exposure to the event under study—the USD-pegged stablecoins that Binance delisted under MiCA—together with the principal compliant coin, USDC, as the control. The large stablecoins not in the treated group are absent for specific reasons rather than oversight: USDT is excluded as the dollar benchmark, and USDC remained compliant and so serves as the control rather than a treated coin. Several other stablecoins were excluded for the following reasons. DAI was delisted from Binance before the sample period and returns no observations on the relevant pairs. Euro-pegged stablecoins such as AEUR and EURI were excluded because their peg is to the euro rather than the dollar, so deviation from one US dollar is not a meaningful measure and they were not subject to the dollar-stablecoin delisting under study. Finally, defunct or non-currency tokens

such as the former Terra USD and the gold-backed PAXG were excluded as outside the relevant contemporary market.

3.3 Descriptive statistics

This subsection describes the data used in the analysis. Table 1 defines each variable, Table 2 reports summary statistics over the two ±7-day event windows, and Table 3 breaks the outcome down by coin within each window. Because the difference-in-differences estimates are computed on the event windows rather than the full sample, the statistics are reported over those windows, each comprising 1,344 coin-hours across fifteen calendar dates.

Table 1 *Variable definitions*

Variable	Definition
Abs. peg deviation	Absolute difference between a coin's hourly closing price and its one-to-one peg, |close − 1|, in basis points (× 10,000). The dependent variable.
Trading volume	Hourly traded volume of the coin on its Binance pair, in quote-currency units.
Treated	One for the delisted coins (FDUSD, TUSD, USDP), zero for the control (USDC).
Post	One for hours from the event onward, zero before; defined per event.
Treated × Post	Interaction; its coefficient is the difference-in-differences treatment effect.
Day	Event-time counter in days relative to the event; used in the parallel-trends test.

Notes: *Variable definitions for the analysis. The sample is an hourly panel of four US-dollar-pegged stablecoins on Binance; all are quoted against USDT (the dollar proxy), and USDT is excluded as it cannot be measured against itself consistently.*

Table 2 *Summary statistics over the ±7-day event windows*

	MiCA window (±7 days)			***Binance window (±7 days)***		
Variable	**Mean**	**SD**	**Max**	**Mean**	**SD**	**Max**
Abs. peg deviation (bps)	9.39	7.93	99.0	12.21	9.26	37.0
Trading volume (millions)	12.04	20.99	159.8	22.27	39.66	451.1
Observations	1,344			1,344		

Notes: *This table reports summary statistics over the two ±7-day estimation windows (each 1,344 coin-hours across 15 calendar dates), which are the samples used in the main difference-in-differences regressions. Absolute peg deviation is in basis points; trading volume is in millions of quote-currency units. Deviation is somewhat higher in the Binance window, reflecting the elevated market stress of late February 2025 that preceded the announcement.*

Table 3 *Absolute peg deviation by coin, within each event window*

	MiCA window (bps)			***Binance window (bps)***		
	30 June 2024			**3 March 2025**		
Coin	**Mean**	**SD**	**Max**	**Mean**	**SD**	**Max**
FDUSD	3.01	2.01	10.0	12.15	4.11	23.0
TUSD	12.27	4.56	22.0	25.90	3.83	37.0
USDP	14.29	11.25	99.0	7.03	3.47	17.0
USDC (control)	7.99	5.03	23.0	3.76	3.68	14.0

Notes: *This table reports absolute peg deviation (in basis points) by coin within each ±7-day event window; each coin contributes 336 hourly observations per window. The treated coins differ in baseline volatility, and the control USDC is tightest in the Binance window, more elevated in the MiCA window. In the Binance window the control's pre-event deviation is low, consistent with the flight toward the compliant coin documented in the results. These cross-coin level differences motivate the coin fixed effects used throughout.*

Two features of the data shape the empirical strategy that follows. First, the stablecoins differ markedly in their baseline level of peg deviation, and they do so differently across the two windows. TUSD is among the most volatile coins in both windows, while FDUSD and USDP vary in their relative position — USDP, for instance, is the highest-deviation coin in the MiCA window but among the lowest in the Binance window. The control coin USDC is the tightest coin in the Binance window, though its deviation is more elevated in the MiCA window, where it exceeds that of FDUSD. What matters for the analysis is not the precise ordering but the fact that the coins occupy persistently different deviation levels; these differences are not of direct interest, but they must be controlled for so that the treatment effect is not confounded by the fact that some coins are simply more volatile than others. They therefore motivate the inclusion of coin fixed effects in every specification, which absorb each coin's average deviation and identify the effect from within-coin changes around the event rather than from cross-coin level differences.

Second, the behaviour of the control coin previews the mechanism examined in the results. In the Binance window USDC's mean deviation is low (3.76 basis points), notably tighter than the treated coins, consistent with a migration of trading activity toward the compliant coin around the announcement. In the MiCA window, by contrast, USDC's deviation is more elevated (7.99 basis

points); this higher pre-event volatility in the control is relevant to the parallel-trends assessment and is revisited in the methodology. Third, trading volume varies widely across coins, as reflected in the high standard deviation of volume in Table 2; this dispersion is exploited in the volume robustness test.

Figure 2 shows that the announcement did not cause the same increase in peg deviations for all treated stablecoins. The stablecoins reacted differently after the announcement. FDUSD remained volatile, TUSD showed lower deviations, and USDP displayed mixed results. The clearest change can be seen for USDC, whose peg deviation decreased after the announcement. This means that the positive difference-in-differences result is mainly caused by a larger gap between the treated stablecoins and the control group, rather than by all treated stablecoins becoming less stable. The reason behind this effect is discussed in the results section.

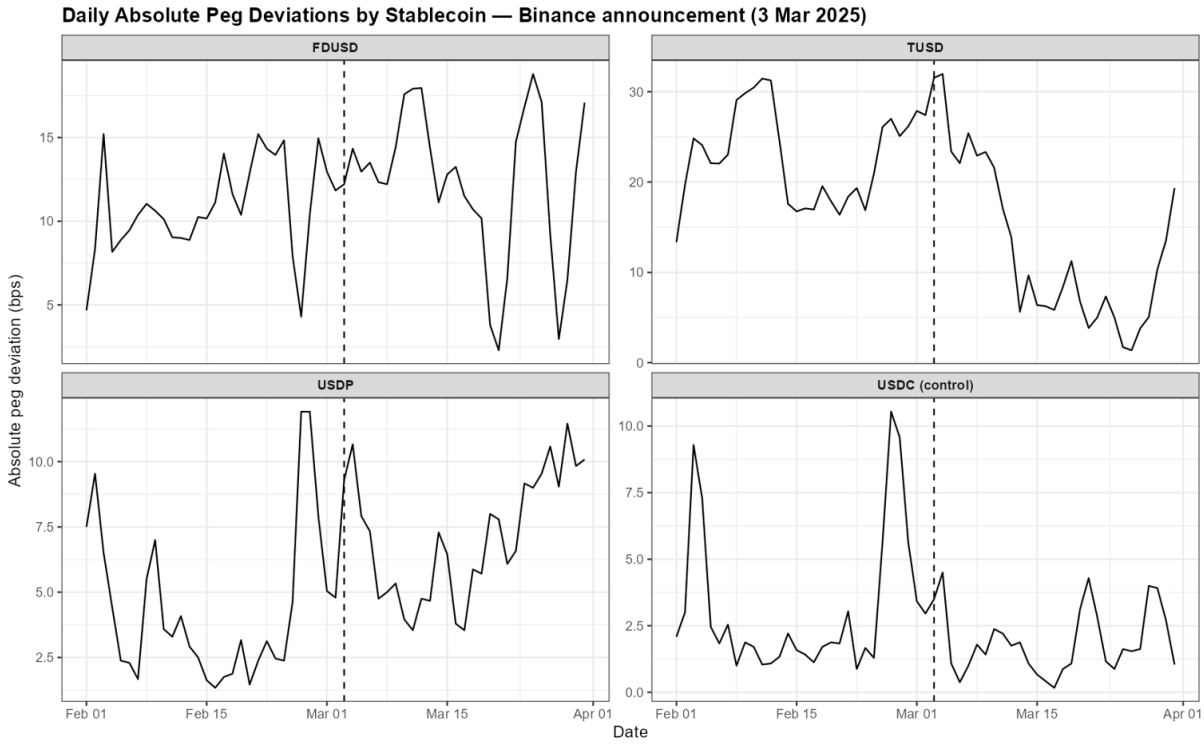


Figure 2. Coin comparison plot

4. Methodology

4.1 Hypothesis development and Main Difference-in-Differences study

The three channels described above information, liquidity, and reallocation together imply a directional prediction for how a MiCA-related exchange restriction should affect peg stability. A delisting announcement conveys new information about the future tradability of the affected coins, reduces the venues and pairs through which they can be traded and arbitraged, and encourages a migration of activity toward the compliant alternative. Each channel points in the same direction: by impairing the market access and liquidity on which arbitrage depends, the restriction should weaken the forces that hold the affected coins at par and widen their short-run deviation from the one-dollar peg. This expectation is consistent with the mechanism literature, which establishes that peg stability depends on the breadth of arbitrage access and on market liquidity, and with evidence that delistings reallocate activity toward compliant coins.

The prediction is relative rather than absolute. Because the compliant control coin, USDC, is exposed to the same market-wide conditions but is not subject to the restriction and may itself benefit from the migration of activity toward it the relevant prediction is that the affected coins' deviation rises relative to USDC, not necessarily that it rises in absolute terms. The difference-in-differences design captures exactly this relative movement. The two events are expected to differ: the broad MiCA application date of 30 June 2024 was anticipated and corresponded to no concrete exchange-level action, so it is not expected to produce a measurable short-run effect and serves as a comparison event; the Binance announcement of 3 March 2025, by contrast, delivered new and specific information about market access on the dominant trading venue, and is expected to produce a significant short-run widening of the affected coins' peg deviation relative to the control. These considerations lead to the central hypothesis tested in this thesis:

H1: Binance's MiCA-related stablecoin restriction announcement widens the short-run gap in absolute peg deviation between the affected coins and the compliant control USDC.

In terms of the specification, H1 implies a positive and statistically significant coefficient β on the Treated × Post interaction for the Binance announcement. The same coefficient is estimated for the 30 June 2024 MiCA application date, which serves as a comparison event; because that date was anticipated and did not correspond to a specific exchange-level restriction, H1 implies no comparable effect there. A significant β for the Binance event together with an insignificant β for the MiCA date would support the interpretation that short-run peg pressure arises when regulation becomes operational through concrete exchange-level restrictions rather than at the broad regulatory milestone.

The baseline specification estimated on the hourly panel is:

$$Dev_{i,t} = \alpha_i + \delta_{d(t)} + \gamma_{h(t)} + \beta(Treated_i \times Post_t) + \varepsilon_{i,t}$$

where $Dev_{i,t}$ is the absolute peg deviation of coin i in hour t, expressed in basis points; α denotes coin fixed effects; δ denotes calendar-date fixed effects; γ denotes hour-of-day fixed effects; Treated equals one for the three delisted coins (FDUSD, TUSD, USDP) and zero for USDC; and Post equals one for hours from the event onward. The coefficient of interest is β, the difference-in-differences estimate. Standard errors are clustered by calendar date throughout.

The dependent variable is absolute peg deviation; the independent variable that identifies the treatment effect is the interaction $Treated_i \times Post_t$. The standalone Treated and Post terms do not appear as separate coefficients in the fully specified model because each is absorbed by a set of fixed effects: Treated is constant within each coin and is therefore absorbed by the coin fixed effects, while Post is constant within each calendar date and is therefore absorbed by the date fixed effects. Only

the interaction, which varies both across coins and across time, survives, and it is precisely this term that measures the differential effect of the event on treated coins.

Standard errors are clustered by calendar date. This choice reflects the structure of the data: because observations are hourly, the unexplained component of peg deviation is strongly correlated across the hours within a given day, as the same liquidity conditions, market sentiment, and news persist intraday. Treating the hourly observations as independent would overstate the amount of independent information in the sample, understating the standard errors and overstating statistical significance. Clustering by date allows the errors within each day to be arbitrarily correlated and bases inference on the number of distinct days rather than the number of hours; it also absorbs common market-wide shocks that affect all coins on the same day. Date is the natural clustering level here, since the dominant dependence is the within-day persistence over time; clustering by coin is not viable given that only four coins are observed. A limitation of this choice is that the tight event windows yield only fifteen date clusters, which is at the lower end of what cluster-robust inference comfortably supports; the stability of the estimate across specifications and its large magnitude relative to its standard error mitigate, but do not eliminate, this concern.

Coin fixed effects assign each coin its own baseline level of deviation, absorbing persistent differences across coins so that a naturally more volatile coin such as TUSD does not bias the estimate. Calendar-date fixed effects absorb common shocks that affect all coins on a given day, most importantly the broad cryptocurrency-market movements that drive deviations up or down for the whole panel at once. Hour-of-day fixed effects, which the hourly frequency makes possible, absorb intraday liquidity patterns: stablecoin liquidity varies over the trading day, and thinner liquidity at certain hours mechanically widens deviation regardless of treatment. By absorbing the average deviation at each hour, these fixed effects ensure the treatment effect is identified from within-hour comparisons of treated and control coins. The stepwise results in Section 5 show that the estimate is

stable as each layer is added, indicating that the hour fixed effects refine precision without being decisive for the result.

4.2 Assessing parallel trends

The parallel-trends assumption is assessed using a linear pre-trend test estimated on data aggregated to daily means within the pre-period. For each event, the pre-period hourly observations are collapsed to one daily mean deviation per coin, and the following regression is estimated:

$$Dev_{i,s} = \alpha_i + \tau \cdot Day_s + \varphi \cdot (Treated_i \times Day_s) + \varepsilon_{i,s}$$

where Day_s is the event-time counter over the pre-period (−7, −6, …, −1), α denotes coin fixed effects, and the dependent variable is the daily mean deviation in basis points. The coefficient τ on the Day term captures the control group's own pre-trend slope, while the coefficient φ on the interaction $Treated_i \times Day_s$ captures the differential pre-trend slope between treated and control coins. It is φ that constitutes the parallel-trends test statistic: an insignificant φ indicates that the treated and control groups were not diverging systematically before the event. Including the Day main effect is essential, because it allows the control group to follow its own trend so that the interaction isolates the difference in slopes rather than the treated group's slope against an assumed-flat control.

Pre-trend tests are known to have limited power and to be sensitive to specification, and conditioning an analysis on the outcome of such a test carries its own costs (Roth, 2022). In the present high-frequency setting a further, practical issue arises: the very large number of observations per coin-day gives a joint hourly pre-trend test so much power that it rejects economically negligible fluctuations. For this reason, the diagnostic is reported on data aggregated to daily means, which is this thesis's own response to that over-rejection rather than a procedure prescribed by any single source. This frequency choice applies only to the pre-trend diagnostic; the main difference-in-

differences model is estimated at hourly frequency. The two are not in conflict, because the main model estimates a level shift while the pre-trend test estimates a slope, which are distinct quantities.

Two robustness exercises support the main estimate. First, the specification is built up stepwise, from a plain difference-in-differences with no fixed effects to the fully saturated model, to show that the coefficient is stable across specifications. Second, trading volume is used to test for heterogeneity by liquidity: the treated coins are split into higher- and lower-volume groups, by their pre event trading volume and the treatment effect is estimated separately for each group, to assess whether the effect is broad based across liquidity levels or concentrated in less liquid coins.

5. Results

5.1 Mechanism

Each coin's peg deviation is measured against the dollar, using USDT as the dollar benchmark, but the treatment effect is identified by comparing the change in the treated coins' deviation with the change in the control coin's deviation, rather than from the treated coins' deviation alone (Sections [3.2] and [4.1]). This comparison structure means the estimated effect can be decomposed into how each group moved around the announcement, which reveals an important nuance in what the difference-in-differences estimate captures. Examination of the group means shows that much of the widening of the treated–control gap arises because the control coin's own deviation fell after the announcement, from approximately 5.6 to 1.9 basis points, as activity shifted toward the compliant coin, while the treated coins' deviation changed comparatively little, from approximately 14.8 to 15.3 basis points. This pattern is also visible in the coin-level series shown earlier in Figure 2: the treated coins, and TUSD in particular, do not worsen in absolute terms after the announcement; rather, it is the sharp improvement in the control coin that drives the widening of the gap. The difference-in-

differences estimate correctly captures this divergence, but the divergence reflects a marked improvement in the control alongside only a modest movement in the treated coins. The result is therefore best described as a relative destabilisation of the treated coins against an improving compliant benchmark, rather than an outright deterioration of the treated coins' pegs in absolute terms.

5.2 Main difference-in-differences estimates

Table 2 and 3 reports the baseline difference-in-differences estimates for the two principal events, each estimated on the plus-or-minus-seven-day hourly window with coin, date, and hour fixed effects and standard errors clustered by date. The contrast between the two events is the central result of the thesis.

Table 4 *Difference-in-differences estimates of the Binance delisting announcement on stablecoin peg deviation*

	(I)	**(II)**	**(III)**	**(IV)**
Constant	5.62***			
	(0.317)			
Treated	9.13***			
	(0.522)			
Post	−3.71***	−3.71*		
	(0.346)	(1.310)		
Treated x Post (bps)	4.26***	4.26***	4.19**	4.34***
	(0.656)	(1.200)	(1.020)	(1.000)
Coin fixed effects	No	Yes	Yes	Yes
Date fixed effects	No	No	Yes	Yes

Hour fixed effects	No	No	No	Yes
Standard errors	Robust	Clustered	Clustered	Clustered
Observations	1,344	1,344	1,344	1,344
R^2	0.289	0.845	0.875	0.877

Notes: This table reports difference-in-differences estimates of the effect of Binance's 3 March 2025 delisting announcement on absolute stablecoin peg deviation, measured in basis points. The treated group comprises FDUSD, TUSD, and USDP; the control is USDC. The sample is the hourly panel over the ±7-day window around the announcement. The coefficient of interest, Treated x Post, is the difference-in-differences estimate. Standard errors, in parentheses, are heteroskedasticity-robust in column (1) and clustered by calendar date in columns (2) and (4). Standalone Treated and Post terms are absorbed by the coin and date fixed effects respectively where included. *** p < 0.001, ** p < 0.01, * p < 0.05, p < 0.10

Table 5 *Difference-in-differences estimates of the MiCA stablecoin rules on peg deviation*

Variable	**(I)**	**(II)**	**(III)**	**(IV)**
Constant	7.41***			
	(0.353)			
Treated	1.37**			
	(0.446)			
Post	1.15*	1.15		
	(0.546)	(2.590)		
Treated × Post(bps)	0.996	0.996	0.996	0.996
	(0.768)	(1.940)	(1.940)	(1.960)
Coin fixed effects	No	Yes	Yes	Yes
Date fixed effects	No	No	Yes	Yes
Hour fixed effects	No	No	No	Yes
Standard errors	Robust	Clustered	Clustered	Clustered
Observations	1,344	1,344	1,344	1,344
R^2	0.026	0.314	0.465	0.470

Notes: This table reports difference-in-differences estimates of the effect of the application of MiCA's stablecoin rules on 30 June 2024 on absolute stablecoin peg deviation, in basis points. The treated group comprises FDUSD, TUSD, and USDP; the control is USDC. The sample is the hourly panel over the ±7-day window around the event. The Treated × Post coefficient is statistically insignificant in every specification, indicating no detectable effect of the regulatory milestone. Standard errors, in parentheses, are heteroskedasticity-robust in column (1) and clustered by calendar date in columns (2) and (4). *** p < 0.001, ** p < 0.01, * p < 0.05, p < 0.10.

Table 2 builds the Binance specification up one layer at a time. The estimate is stable across all four columns, moving only between +4.19 and +4.34 basis points and remaining significant at the one-percent level throughout. This stability indicates that the result is not an artefact of any particular set of controls; the fixed effects sharpen precision but do not change the magnitude of the effect.

In Table 3 MiCA regulatory milestone produced no detectable effect. The estimated coefficient of +0.996 basis points is small relative to average deviation levels, and its confidence interval comfortably spans zero ($p = 0.61$). On the formal application date of MiCA's stablecoin rules, treated coins did not deviate from par any differently than the compliant control. This is consistent with a market that had anticipated the regulation, the substance of which had been known since MiCA's adoption in 2023, and had priced its consequences in advance.

The Binance announcement, by contrast, produced a statistically significant widening of peg deviation. Treated coins' absolute deviation rose by 4.34 basis points relative to USDC following the announcement, an estimate that is precisely determined ($p < 0.001$) and economically meaningful against baseline deviations of single to low-double-digit basis points. The announcement of a concrete delisting, even though it flowed from the same regulation, delivered new information about the future tradability of the affected coins and was associated with immediate widening of their pegs relative to the compliant benchmark.

5.3 Parallel-trends test

Table 4 reports the parallel-trends test for both events. The coefficient of interest is the differential pre-trend slope (*Treated × Day*), which measures whether treated and control coins were diverging before the event; an insignificant coefficient supports the assumption.

Table 6 *Linear pre-trend test of the parallel-trends assumption*

Variable	**MiCA event**	**Binance event**
	(30 Jun 2024)	***(3 Mar 2025)***
Treated × Day	−1.432*	0.883
	(0.644)	(0.891)
Day	1.580**	−0.516
	(0.499)	(0.810)
Coin fixed effects	Yes	Yes
Standard errors	Robust	Robust
Observations	28	28
R^2	0.742	0.870

Notes: This table reports the linear pre-trend test used to assess the parallel-trends assumption, estimated on pre-event data aggregated to daily means (seven pre-event days for each of four coins). The coefficient of interest, Treated × Day, is the differential pre-trend slope between treated and control coins, in basis points per day; the Day coefficient is the control group's own pre-trend slope. An insignificant Treated × Day coefficient supports parallel trends. For Binance the differential slope is insignificant ($p = 0.32$), supporting the assumption. For MiCA the differential slope is significant at the 5% level ($p = 0.03$); because the MiCA treatment effect is itself statistically insignificant, this pre-trend does not affect the substantive conclusion. Heteroskedasticity-robust standard errors in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, $p < 0.10$.

For the Binance announcement, the differential pre-trend slope is small and clearly insignificant (+0.88 basis points per day, $p = 0.32$), and the control coin shows no significant pre-trend of its own. Parallel trends are therefore well supported for the event on which the significant treatment effect is found. For the MiCA event, the differential slope is larger in magnitude (−1.43 basis points per day) and is significant at the 5% level ($p = 0.03$), so the parallel-trends assumption is rejected for this event. The significant positive coefficient on the Day term (+1.58 basis points per day) shows that the control coin was itself trending upward immediately before the regulatory date, reflecting elevated USDC volatility in that window. This pre-trend violation does not, however, threaten the substantive conclusion: a violation of parallel trends can bias an estimated effect, but it cannot manufacture significance where there is none, and the MiCA treatment effect is statistically insignificant. The MiCA finding is therefore best read as the absence of any detectable short-run effect at the regulatory

date, a conclusion that holds regardless of the pre-trend, while clean identification of a significant effect is found only for the Binance announcement. This pattern is also visible in the event-study estimates shown in Figure 3. For the Binance announcement, the pre-event coefficients are flat and close to zero, providing visual support for parallel trends, consistent with the formal test in Table 4. The corresponding MiCA event-study figure, reported in the Appendix, shows more scattered pre-event coefficients, consistent with the rejected pre-trend test for that event.

5.4 Dynamic path of the announcement effect

Beyond the pre-event period, the event-study estimates in Figure 3 also trace the dynamic path of the Binance effect, which is sharply concentrated at the announcement and absorbed within two days. The difference-in-differences coefficient peaks on the announcement day at approximately +5.4 basis points and remains elevated on the following day (+4.2 basis points), before falling sharply to near zero from the second day onward. This pattern, a pronounced spike concentrated in the first 48 hours, followed by rapid absorption is the signature of a short-run information shock that the market processes quickly, rather than a permanent level shift. Figure 3 does not show absolute peg deviation directly it shows estimated coefficients (differences from reference day) and can be negative even though absolute deviation is non negative and each coefficient measures the difference between the treated–control gap on that day and the gap on a chosen reference day. Because these are differences relative to a baseline, they can be positive or negative: a negative coefficient simply means the treated–control gap on that day was smaller than it was on the reference day, not that any coin's deviation was below zero. The MiCA event study figure is reported in the appendix.

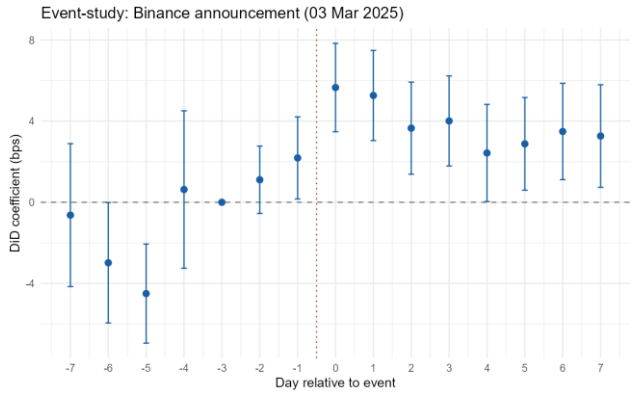


Figure 3. Daily event-study coefficients.no discontinuity at the event: Binance (3 March 2025), a clear announcement-day jump that decays over the week.

5.5 Volume robustness

To assess whether the peg effect is concentrated in less liquid coins, trading volume is used to test for heterogeneity by liquidity. Splitting the treated coins by pre-event volume, the effect is positive and significant for both the higher-volume coins (FDUSD and TUSD, +4.72 basis points, $p = 0.002$) and the lower-volume coin (USDP, +3.58 basis points, $p < 0.001$), and a continuous interaction of the treatment with log pre-event volume is statistically insignificant (+0.39 basis points per log unit, $p = 0.14$). The effect is therefore broadly present across liquidity levels rather than concentrated in a single illiquid coin, although with three treated coins the heterogeneity analysis is necessarily coarser than it would be with a larger cross-section. The volume robustness results should be interpreted as evidence on heterogeneity rather than as a separate main effect. The high- and low-volume group specification shows that the Binance announcement effect is positive and statistically significant for both liquidity groups. This indicates that the result is not driven solely by one low-liquidity stablecoin and that they are distributed across the treated coins.

Table 4 *Heterogeneity of the announcement effect by trading volume (liquidity)*

Variable	**High/Low volume groups**
	(1)
Treated × Post × High volume (bps)	4.70**
	(1.500)
Treated × Post × Low volume (bps)	3.60***
	(0.722)
Coin / Date / Hour fixed effects	Yes
Standard errors	Clustered
Observations	1,344
R^2	0.878

Notes: This table reports the heterogeneity of the Binance announcement effect by trading volume, in basis points, over the ±7-day hourly window. In column (1) the treated coins are split into higher-volume (FDUSD, TUSD) and lower-volume (USDP) groups by pre-event volume. Both groups show a positive, significant effect of similar magnitude indicating the effect is broadly present across liquidity levels rather than concentrated in a single illiquid coin. All specifications include coin, date, and hour fixed effects; standard errors clustered by date in parentheses. *** p < 0.001, ** p < 0.01, * p < 0.05, p < 0.10.

6. Discussion

This section discusses the main limitations of the analysis and the extensions they motivate. For each, it considers how the limitations bear on the results, what the existing literature suggests would change if it were addressed and what future research work could do.

The foremost limitation concerns the dependence on a single exchange. The main result studies the announcement of 3 March 2025, during which the affected coins were still trading normally on Binance. Their prices over the announcement window are therefore valid and form the basis of analysis. The limitation arises only for the effective delisting of 31 March 2025, once the coins were removed and Binance prices became stale. So, the actual removal which is distinct from the announcement cannot be studied on this venue and that event is reported inconclusive and for this reason the clean result is confined to the announcement. Measuring the affected coins on the

exchanges where they continued to trade after the delisting would be required to identify that effect, and the cross-venue liquidity reallocation documented by Diop and Chevallier (2026) suggests it could be substantial, since the removal of a dominant venue would force a larger reallocation than its announcement alone. A multi-venue design is therefore the most immediate extension, and would allow the effect of the actual delisting, rather than only its announcement, to be identified.

A second limitation is that the outcome is measured against USDT as a dollar proxy rather than against US dollar because Binance does not quote these coins against fiat dollars. Measuring stablecoins against USDT is common practice in studies using exchange-level data, since USDT is the dominant unit of account on most venues; its limitation is that the absolute level of deviation might differ against a true dollar reference. The effect on the estimated treatment effect is likely limited because all coins share the same USDT benchmark, so any common movement in USDT is differenced out. Future work drawing on off-exchange or aggregated dollar prices could be one way to solve this.

The third limitation is statistical. The tight event windows yield only fifteen calendar-date clusters, at the lower end of the range in which cluster robust standard errors can over-reject. In the present setting, the stability of the estimate across specifications and its large magnitude relative to its standard error mitigate, but do not eliminate, the concern. This is a recognised concern in the difference-in-differences literature (Cameron, Gelbach and Miller, 2008). Future applications could address it directly, either by pooling a larger number of delisting events to raise the cluster count or use bootstrap-based inference for few clusters as proposed by Cameron, Gelbach and Miller (2008).

A fourth limitation concerns the parallel-trends assumption, which cannot be directly verified and whose pre-trend diagnostics are themselves imperfect—a limitation inherent to the difference-in-differences design rather than specific to this study (Roth, 2022). The assumption is supported for the Binance event but rejected for the MiCA event, owing to elevated pre-event volatility in the control coin. Because the MiCA effect is itself statistically insignificant, a pre-trend violation cannot

account for the result such a violation can bias an estimated effect but cannot manufacture significance where there is none so the conclusion is unaffected. A cleaner pre-period, or an alternative compliant control less prone to idiosyncratic volatility around the regulatory date, would nonetheless strengthen identification for the MiCA event, and future work with a wider set of compliant controls could provide this.

Finally, with only three treated coins, the analysis of liquidity heterogeneity is necessarily coarse, resting on a single coin in lower-volume group. The arbitrage-concentration evidence of Ma, Zeng and Zhang (2025) suggests that lower liquidity coins, supported by fewer arbitragers should be more sensitive to a restriction; with three coins this prediction cannot be tested with precision and the volume results should be read as indicative rather than definitive. Studies of the broader stablecoin market work with far larger cross-sections, and extending the present framework to the delisting actions of other exchanges around the same regulatory deadlines would yield a comparable cross-section, allowing the relationship between liquidity and peg sensitivity to be identified more finely while also testing whether the pattern documented here generalises beyond Binance. Two further extensions follow naturally: applying the framework to euro-pegged stablecoins, which MiCA treats under distinct provisions, would permit a comparison of the regulation's impact across peg currencies; and a microstructure analysis using order-book data would offer a richer picture of how delistings affect liquidity and price formation than closing-price deviations alone can provide.

7. Conclusion

This thesis asked whether MiCA and the exchange delistings it triggered increased short-run deviation from par among affected stablecoins, and whether the effect differed between the regulatory milestone and a concrete, exchange-specific restriction announcement by Binance. Using hourly Binance data and a difference-in-differences event-study design, with the compliant coin

USDC as a control, it studied two events: the application of MiCA's stablecoin rules on 30 June 2024, and Binance's delisting announcement of 3 March 2025, during which the affected coins were still trading.

The two events yield a sharp contrast, which is the central finding of the thesis. The MiCA announcement date had no detectable effect on peg deviation. The Binance announcement, by contrast was associated with a statistically significant short run widening of peg deviation among affected coins relative to the control; an effect of roughly four basis points that peaked at the announcement and was absorbed within two days, remained stable across specifications and did not depend on coin liquidity. These results support a clear interpretation. The two events differ not in regulation behind them, which is the same in both cases but in the kind of information each conveyed. Because MiCA has been adopted in 2023 and its requirements were widely known, an anticipated milestone conveyed little new information on its effective date, which is consistent with the absence of any measurable effect. Although the Binance announcement followed from the same regulation, it was a discrete and imperfectly anticipated action by a dominant venue that conveyed concrete information about the future tradability of the affected coins and it moved their pegs. The central contribution is therefore to distinguish the effect of anticipated regulation from that of its concrete enforcement in this case, the market impact of MiCA was realised not when the rule formally applied, but when a major exchange announced it would act upon it.

A closer reading of the effect refines this conclusion. The widening of the treated-control gap arose substantially because the compliant control coin’s own deviation improved as activity migrated toward it, while the affected coins’ deviation changed comparatively little. The result is therefore best understood as a relative destabilisation of the affected coins against an improving compliant benchmark, rather than an outright deterioration of their pegs. A reading consistent with the role of compliance as a determinant of market access under MiCA (Santner and Taudes, 2025) and with the reallocation of liquidity toward the compliant coin following a delisting (Diop and Chevallier, 2026).

These findings carry several implications. For the study of regulation and markets, they illustrate, in a new setting, that the market impact of a rule is realised not necessarily at its formal entry into force but at the moments of concrete, imperfectly anticipated enforcement. For policymakers and regulators, they suggest that the disruptive potential of stablecoin regulation is concentrated in the enforcement actions that follow it, particularly the announcement and removal of trading venues rather than in the passage or application of the rule itself, so that the orderly management of delistings and the availability of compliant alternatives may matter more for short run stability than the timing of regulation. For market participants and issuers, the asymmetry between the affected coins and the compliant control is instructive i.e. the coin that secured authorisation not only avoided destabilisation but appears to have benefited from the migration of activity toward it, underscoring the competitive consequences of regulatory compliance in stablecoin market.

Subject to the limitations of single venue measurement and a short identification window around the announcement, this thesis offers the distinction between a regulatory rule and its concrete enforcement as a lens through which the short tun market consequences of crypto asset regulation may be more generally understood.

References

ABP. (2026). Press release of ABP Quarterly Report Q4 2025. https://www.abp.nl/english/press-releases/2026/january/quarterly-report-q4-2025

Ahmed, R., Aldasoro, I., & Duley, C. (2025). Public information and stablecoin runs (BIS Working Papers No. 1164). Bank for International Settlements. https://www.bis.org/publ/work1164.pdf

Lyons, R. K., & Viswanath-Natraj, G. (2023). What keeps stablecoins stable? *Journal of International Money and Finance, 131*, 102777. https://doi.org/10.1016/j.jimonfin.2022.102777

Binance. (2025, March 3). Binance will delist non-MiCA compliant stablecoin trading pairs for EEA users on 2025-03-31. https://www.binance.com/en/support/announcement/detail/bcaa1f68d6a6450099056ff694ad6c46

DefiLlama. (n.d.). Stablecoin market cap chart, supply and peg data. Retrieved June 18, 2026, from https://defillama.com/stablecoins

Diop, P. O., & Chevallier, J. (2026). Binance USD delisting and stablecoins repercussions: A local projections approach. Econometrics, 14(1), 6. https://doi.org/10.3390/econometrics14010006

European Parliament and Council of the European Union. (2023). Regulation (EU) 2023/1114 of the European Parliament and of the Council of 31 May 2023 on markets in crypto-assets, and amending Regulations (EU) No 1093/2010 and (EU) No 1095/2010 and Directives 2013/36/EU and (EU) 2019/1937. Official Journal of the European Union. https://eur-lex.europa.eu/eli/reg/2023/1114/oj/eng

A. Colin Cameron & Jonah B. Gelbach & Douglas L. Miller, 2008. "**Bootstrap-Based Improvements for Inference with Clustered Errors**," The Review of Economics and Statistics, MIT Press, vol. 90(3), pages 414-427, August.

Ma, Y., Zeng, Y., & Zhang, A. L. (2025). Stablecoin runs and the centralization of arbitrage (NBER Working Paper No. 33882). National Bureau of Economic Research. https://doi.org/10.3386/w33882

Roth, J. (2022). Pretest with caution: Event-study estimates after testing for parallel trends. American Economic Review: Insights, 4(3), 305–322. https://doi.org/10.1257/aeri.20210236

Santner, M., & Taudes, A. (2025). The impact of MiCAR on the euro stablecoin market (Working Paper Series / Institute for Cryptoeconomics / Interdisciplinary Research). WU Vienna University of Economics and Business. https://doi.org/10.57938/dccaf62b-30f6-4824-b4a0-2a240b081ce2

Marcin Wątorek, Stanisław Drożdż, Jarosław Kwapień, Ludovico Minati, Paweł Oświęcimka, Marek Stanuszek,Multiscale characteristics of the emerging global cryptocurrency market, 2021.Pages 1-82, ISSN 0370-1573, https://doi.org/10.1016/j.physrep.2020.10.005.

Gadzinski, G., Castello, A., Liuzzi, V., & Sargenti, P. (2024). Break a peg! A study of stablecoin co-instability. *International Review of Financial Analysis, 96*, 103608. https://doi.org/10.1016/j.irfa.2024.103608

De Blasis, R., Galati, L., Webb, A. *et al.* Intelligent design: stablecoins (in)stability and collateral during market turbulence. *Financ Innov* **9**, 85 (2023). https://doi.org/10.1186/s40854-023-00492-4

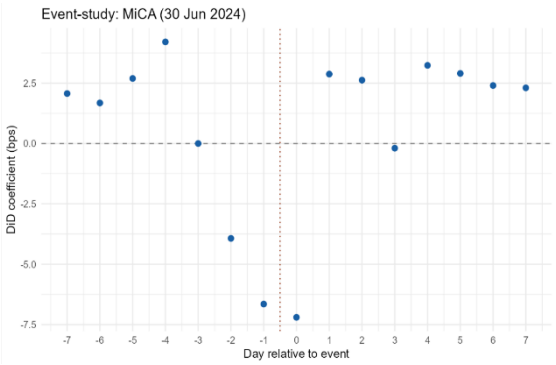
David Vidal-Tomás, Centralized exchanges & proof-of-solvency: The guardians of trust, Journal of International Financial Markets, Institutions and Money, Volume 103,2025,102183, ISSN 1042-4431, https://doi.org/10.1016/j.intfin.2025.102183.

Cho-Hoi Hui, Andrew Wong, Chi-Fai Lo, Stablecoin price dynamics under a peg-stabilising mechanism, Journal of International Money and Finance, Volume 152,2025, 103280, ISSN 0261-5606, https://doi.org/10.1016/j.jimonfin.2025.103280.

Appendix

Appendix A. Event-study figures

Figure A1 shows the daily event-study coefficients for both events, with each point the difference-in-differences coefficient for one day relative to the event and vertical bars denoting 95 percent confidence intervals; the reference day is normalised to zero. The contrast is the visual counterpart of the main result: there is no discontinuity at the MiCA date, whereas the Binance announcement shows a clear upward jump at the event that decays over the following days.


Event-study: MiCA (30 Jun 2024)
DiD coefficient (bps)
2.5
0.0
-2.5
-5.0
-7.5
-7
-6
-5
-4
-3
-2
-1
0
1
2
3
4
5
6
7
Day relative to event